

RESEARCH ARTICLE

Use of evidence for decision-making by conservation practitioners in the illegal wildlife trade

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Abstract

1. There are calls to ground policies aimed at addressing the illegal wildlife trade (IWT), and biodiversity conservation more generally, on the best available evidence. However, evidence on IWT can be highly uncertain and difficult to obtain due to the illegal nature of the trade. Even when the evidence exists, there are numerous barriers to its uptake by decision makers, pertaining to the evidence itself and to the characteristics and contexts of those using it.
2. The surfacing of the illegal trade in jaguars *Panthera onca* provides an example of how evidence is, and is not, used for decision-making on IWT. We interviewed 38 conservation practitioners in Belize, Guatemala and Honduras, who had knowledge about, or experience dealing with, illegal jaguar trade. Interviewees described their information sources and decision-making processes, and explicitly and implicitly prioritized jaguar trade evidence, based on attributes like the evidence's source, the scale and purpose of the trade, its temporal and spatial dimensions and the nationality of offenders.
3. Even though interviewees stated that they use scientific evidence to make decisions, they gave more weight to evidence involving foreign actors and commercial purposes than local and non-commercial ones, paying less attention to the potential impact on jaguars or the source of the information. They were also more inclined to favour events that were spatially and temporally closer to their own reality. Our results show that the interpretation and uptake of evidence are subject to contextual constraints and personal biases, which are common across fields and sectors, even amongst experienced decision makers.
4. We propose an approach for evaluating evidence and informing decision-making within IWT and biodiversity conservation. Our approach aims to guide conservation decision makers and practitioners to assess the relevance and uncertainty of the evidence, to recognize their own interpretation biases, to identify the actions that are appropriate based on the evidence and to improve the transparency of their decisions. It can also guide evidence 'producers' to develop evidence that is more aligned to conservation policy and practice.

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5. This approach can contribute towards more evidence-based practice within the field of biodiversity conservation, with applications to IWT and beyond.

1 | INTRODUCTION

Biodiversity loss is one of the most pressing environmental issues of our time. The unprecedented anthropogenic decline in biodiversity across the world threatens the life support systems on which humans depend (WWF, 2020). As a mission-driven discipline tasked with addressing this impending crisis, the role of conservation science extends beyond monitoring this decline, to informing conservation policies and management strategies and having a practical impact on the preservation of biodiversity and on the improvement of human well-being (Kareiva & Marvier, 2012; Toomey et al., 2017). However, a growing body of literature has highlighted the lack of evidence-based conservation, and associated it with the continued decline in biodiversity (Cook et al., 2013; Sutherland et al., 2004; Sutherland & Wordley, 2017).

Studies on the use of evidence in conservation have identified multiple challenges in navigating the space between research, decision-making and implementation in conservation (Maas et al., 2019; Toomey et al., 2017; Wright et al., 2020). The first hurdle arises from the multiple definitions of 'evidence', and a lack of consensus on what counts as 'good evidence' for conservation policies (Oliver & Cairney, 2019; Salafsky et al., 2019). Depending on the context, evidence can refer to a physical item, a set of accumulated facts about a situation, an assessment of the validity of the facts, a body of relevant theory or confidence that an assertion about a situation is true (Salafsky et al., 2019). In the case of conservation science and practice, Salafsky et al. (2019) defined evidence as 'relevant information used to assess one or more hypotheses related to a question of interest', with 'information' varying in complexity and quality, from basic, unfiltered data or raw observations, to a critically appraised evidence base (e.g. systematic reviews). However, the lack of a widely accepted definition of evidence in conservation science and practice poses challenges to producing standard guidance on the use of evidence (Salafsky et al., 2019).

Other challenges have to do with the availability and characteristics of the evidence itself. To be relevant, the evidence must match the temporal and geographical scales at which decisions are made and meet both scientific (e.g. generalizability and methodological rigour) and social quality standards (e.g. reliability and legitimacy) to be considered trustworthy (Cash et al., 2003; Cooke et al., 2017; McNie et al., 2016; Oliver & Cairney, 2019). Further challenges relate to the process of communicating science, from one-way knowledge flows from science to practice, to more dynamic participatory processes that include different voices, and which address power imbalances through mediation (Bertuol-Garcia et al., 2018; Nguyen et al., 2017). How the evidence is framed or packaged to achieve a desired result also influences its communication and uptake (Elliott & Nisbet, 2018; Nutley et al., 2012). Decision

makers are more likely to react to information that is vividly framed, plausible, emotionally charged and well publicized, and evidence which they perceive as being personally relevant or close to their own reality (Newell et al., 2014; Spence & Pidgeon, 2010; Trope & Liberman, 2010; Witting, 2017). This also depends on the individual characteristics of those involved in knowledge exchange, such as their environment, experience, values, education and learning style (e.g. heuristic or intuitive vs. analytical processing, Witting, 2017); and on the wider governance structures and socio-political contexts of decision-making, including the incentive systems motivating scientists and decision makers to engage with each other (Walsh et al., 2019; Wright et al., 2020). While these (and other) challenges to evidence-based decision-making have been repeatedly identified in the conservation literature and in other disciplines (e.g. sociology, health care, psychology, science and technology, policy sciences), and many solutions have been proposed, implementation challenges (e.g. lack of resources, support and rewards) and lack of integration across disciplines and sectors continue to hamper evidence-based conservation and lead to research wastage (Maas et al., 2019; Oliver & Boaz, 2019; Oliver & Cairney, 2019; Rose et al., 2018).

The illegal wildlife trade (IWT) is a key driver of biodiversity decline, and it has great policy resonance worldwide, drawing considerable attention and financial resources ('t Sas-Rolfes et al., 2019; Massé & Margulies, 2020). However, like other topics in conservation, IWT is affected by the multiple challenges to evidence-based conservation, particularly the lack of evidence surrounding its prevalence, characteristics and drivers (Milner-Gulland et al., 2018; Symes et al., 2018). Due to its covert nature, most of the available information on IWT is restricted to seizure events reported by national authorities or documented by the media (UNODC, 2020). Seizure data on its own cannot adequately represent the magnitude or trends in IWT, being subject to numerous biases including unknown proportions of seized items; unidentified seizure rates; varying enforcement effort, effectiveness and reporting across countries; taxonomic and product biases; and lack of accessibility to the data, among others (Symes et al., 2018; Underwood et al., 2013; UNODC, 2020). In the absence of robust estimates and context-specific analyses of the characteristics of IWT for many illegally traded species, decision-making on IWT is often subject to misinformation, lobbying and emotionally and geopolitically appealing narratives about its potential links with terrorism, national security and foreign demand (Duffy, 2014; Massé & Margulies, 2020). These dominant narratives have perpetuated stereotypical representations of the actors involved in IWT, particularly of marginalized rural communities in source areas and of consumers of Asian descent, making ungrounded generalizations and failing to account for their motivations and cultural specificities (Margulies et al., 2019). As these narratives dominate

the IWT policy space, often facilitated by the media, there could be unintended consequences for the wildlife concerned, resulting from a lack of nuanced interventions, the criminalization of local livelihoods and the neglect of key organisms with respect to policy attention and funding (Duffy, 2014; Milner-Gulland et al., 2018).

The recent surfacing of the illegal trade in jaguar *Panthera onca* body parts is an example of how evidence is, and is not, used for decision-making on IWT. While the trade in jaguar body parts is a longstanding practice, rooted in the cultural traditions of numerous indigenous and mixed-ethnicity societies throughout Latin America (Saunders, 1998), it has recently gained particular visibility following the discovery of its links with demand from Chinese wildlife markets (Nunez & Aliaga-Rossel, 2017). An apparent increase in seizures of jaguar body parts over the past decade, a portion of which involved China as a destination country or traders of Chinese descent, sounded the alarm that a new market may be emerging for jaguars (Fraser, 2018). Although investigations on the drivers of jaguar trade are starting to emerge (Arias et al., 2020, 2021; Morcatty et al., 2020), the lack of timely and robust evidence beyond seizures allowed for the spread of unsubstantiated assumptions about the trade. This included fears that jaguars may be replacing tigers in Traditional Chinese Medicine (TCM), and that jaguar trade may be driven by Chinese mafias and China's corporate investments in Latin America (e.g. Romo, 2020). Regardless of the continued uncertainty around these narratives and the magnitude of the trade, the combination of a highly charismatic species, threatened by an archetypal 'Asian Super Consumer' (Margulies et al., 2019), employing cruel and criminal tactics (e.g. Romo, 2020; WAP, 2018), captured the attention of the conservation community. This resulted in the mobilization of substantial resources and conservation efforts to address jaguar trade, and the positioning of the jaguar as the emblem of Latin America's fight against IWT in 2019 (High Level Conference on IWT in the Americas, 2019). While these efforts are a clear manifestation of the precautionary principle in conservation, which aims to anticipate, monitor, prevent and mitigate potential threats to biodiversity in cases of uncertainty (Cooney & Dickson, 2012), the decision to prioritize international jaguar trade without understanding its drivers or impacts on the species comes with trade-offs that must be considered.

Here, we explore the use of evidence by conservation practitioners working to address illegal jaguar trade, and IWT more broadly, in Mesoamerica. We sought to understand how evidence is used to make decisions on IWT, the types of IWT evidence that are commonly consulted, how different types of evidence are prioritized and what challenges exist in transforming evidence to action on the ground. We conducted semistructured interviews with key conservation actors, discussing their views on, and experiences with, evidence and decision-making on IWT and jaguars specifically. Additionally, we implemented a jaguar trade evidence evaluation exercise, which required practitioners to implicitly and explicitly assess certain attributes in illegal jaguar trade evidence, including its source, the temporal and spatial setting of the events described, the actors and purposes of the trade and its scale. Based on construal

level theory (CLT, Trope & Liberman, 2010), a contemporary theory in social psychology with applications in decision-making science, we hypothesized that these attributes would affect the perceived 'psychological distance' of the evidence, or in other words, how removed the evidence is from participants' personal experience. The temporal, spatial, social and hypotheticality dimensions of psychological distance impact how people represent events (concretely or abstractly), ultimately influencing their perceptions, reactions and behaviours towards information (Trope et al., 2007). We hypothesized that participants would prioritize events and information:

1. Involving local or current events that are closer to their reality and more feasible for them to personally respond to;
2. Involving foreign actors, commercial drivers and large scales of trade, assuming they may involve a higher risk due to their recent, uncertain and unfamiliar nature;
3. Originating from official sources that increase the plausibility and trustworthiness of the information.

Our goals are threefold: to gain a deeper understanding of evidence use and decision-making by conservation practitioners working to address IWT on the ground; to determine whether there are any intrinsic biases in evaluating and prioritizing IWT evidence; and based on these insights, to propose an approach for evaluating and prioritizing IWT evidence of varying types and qualities to support evidence-based conservation actions.

2 | METHODS

We interviewed 38 conservation practitioners in Belize ($n = 15$), Guatemala ($n = 21$) and Honduras ($n = 2$) as part of a larger study on the illegal trade in jaguar body parts (see Arias et al., 2020). We selected these countries due to anecdotal and official reports of jaguar trade, and their important role in IWT more broadly, with high-profile cases involving scarlet macaws *Ara macao* and rosewood *Dalbergia* spp (Gluszek et al., 2020).

Interviewees were either actively or recently (within the past 5 years) involved in the response to IWT in their countries, including jaguar trade. They included jaguar scientists ($n = 3$), wildlife rescue centre directors ($n = 4$), NGO staff involved in jaguar conservation ($n = 16$), protected area managers ($n = 5$), as well as government and law enforcement representatives ($n = 10$), from 19 different institutions. We used a three-step approach to identify potential interviewees using both purposive and snowball sampling. First, we listed individuals involved in IWT cases and jaguar conservation initiatives in our study countries, based on manual key word internet searches, focusing on academic publications, grey literature, news reports as well as institutional websites (belonging to national and regional governments and NGOs). Second, we reviewed this initial list of contacts with our local partners, WCS Mesoamerica, to ensure that we had a comprehensive list of participants working on these topics. Third, we approached 48 individuals in the list by email and invited them to

suggest additional people within their network who may fit our criteria, adding five people to the list through snowball sampling. Out of 53 people contacted, 41 agreed to participate in the interview and 38 completed the full interview (the sample size of this study). Interviews were in person ($n = 35$) or via Skype ($n = 3$), and were conducted by the lead author (M.A.) in September–November 2018. The portion of the interview on which this paper is based took 15–20 min to complete, and followed a 45-min discussion about the characteristics of illegal jaguar trade (reported in Arias et al., 2020). Interviews were audio-recorded upon obtaining written (for in-person interviews) or oral consent (for Skype interviews) from participants, but participant identity is kept confidential. The study's interviewing, informed consent and confidentiality protocols were reviewed and approved by the Central University Research Ethics Committee of Oxford University (Reference number: R59134/RE001).

We asked interviewees a standardized set of semistructured questions about how they make decisions on what topics and activities to prioritize within their IWT-related work, their information sources, their responses to evidence on jaguar trade and IWT and the challenges that they face when using evidence. Following this discussion, interviewees participated in a short exercise in which they were asked to read 10 cards containing short extracts (1–3 sentences) of different types of jaguar trade evidence, and to rank them in terms of their priority for conservation action (Table 1). The cards were presented to them in a random order, and the extracts were designed so that they varied in source (government, media, acquaintances and research), nationality of the perpetrator (national or foreign), drivers of the trade (commercial or non-commercial), scale (quantitative or undefined), location (domestic or foreign country) and time (current or past). Due to interview time constraints, the extracts were deliberately short, and they may not necessarily represent the actual type of 'evidence' that conservation practitioners normally use to make decisions on IWT. Nevertheless, they represent evidence leads that fit Salafsky et al.'s (2019) definition of evidence as any 'relevant information used to assess a hypothesis related to a question of interest', and which we assumed would prompt a reaction or opinion from our interviewees. Moreover, the extract attributes were inspired by real events and realistic IWT and illegal jaguar trade scenarios, meant to resemble the types of evidence that conservation practitioners are commonly exposed to.

Our extract design was inspired by multicriteria decision analysis (e.g. Burgman, 2005) and choice experiments (e.g. Hinsley et al., 2015), which evaluate outcome decisions or product choices based on a structured and systematic comparison of their attributes. These methods commonly employ orthogonal designs and repeated pairwise comparisons to isolate the effects of individual attributes and to reach optimal attribute combinations that reflect the preferences of decision makers or consumers. Our card design followed a similar approach, but we did not present our interviewees with all potential attribute combinations to reduce the cognitive demand on participants, and because our sample size of relevant interviewees with key expertise on jaguars and IWT in these countries was inevitably small. We did not emphasize a balanced design, placing more

weight on the hypotheses that were of most interest to us (which pertained to the source, trader nationality and purpose attributes), and because our intention was to use the cards as props to encourage a discussion about evidence, rather than to reach an optimal card arrangement. Once interviewees organized the cards, we took photographs of the physical (or virtual, for Skype interviews) card arrangement and asked interviewees to explain how they prioritized the cards, without giving them any details about the attributes that were encoded in them. Follow-up questions focused on eliciting additional details about the prioritization process, by comparing sets of cards that shared similar attributes but differed in others, and later by openly asking them about their thoughts on the different evidence attributes found in the cards.

We transcribed the interviews and coded the answers to the semistructured questions of the interview into question–answer categories and subcategories (or themes) using NVivo qualitative data analysis software (QSR International Pty Ltd. Version 12 Pro, 2018) and following the guidelines for thematic analysis (Newing, 2011). Four randomly selected interviews were also coded by another researcher in the team to test the reliability of our coding, reaching a Cohen's Kappa of 0.86, or excellent agreement.

We analysed the card ranking exercise following two different approaches. In the first one, we identified the attributes that interviewees described as being important when explaining their card ranking process without being told about the attributes encoded in the cards. Drawing from the literature on policy learning, social psychology and consumer explicit and implicit preferences (Friese et al., 2006; Perugini, 2005; Witting, 2017), we considered these attributes as the 'explicit' criteria employed by interviewees for prioritizing evidence, which are a manifestation of their deliberate and conscious 'analytical processing' of the evidence and its value. In the second approach, we evaluated the actual physical arrangement of the cards, recording the order given to each card (from 1–10, with 10 being the highest priority) from photographs that were taken of the card organization. We considered this ordering as the 'implicit' criteria employed by interviewees for prioritizing evidence, representing their spontaneous reactions to the evidence or their heuristic (or intuitive) processing of the information (Witting, 2017). We used card rank as the response variable, and the binary or multi-level factor evidence attributes (source, nationality, purpose, scale, location, time) as predictors in a cumulative link mixed model (or ordered logistic generalized linear model) with interviewee ID as the grouping variable, using the function 'clmm' in package *ORDINAL* (Christensen, 2019) in R version 3.6.2 (R Core Team, 2019). We calculated 'Wald' confidence intervals. Model variables were tested for association prior to inclusion in the model applying Cramer's V through the package *DescTools* in R (Andri et al., 2021). Medium and low levels of association (<0.7) were found, allowing us to keep all attributes of interest in the model. However, it should be noted that the attributes in the cards are not fully independent and that our sample size is small, which means that the associations between evidence attributes and prioritization are indicative only, and meant to be examined in relation to the deeper qualitative

TABLE 1 Jaguar trade evidence cards and their attributes. Nationality refers to the citizenship or cultural background of those involved in jaguar trade relative to the country where the events took place. Source refers to the origin of the information on trade. Scale refers to whether the examples of jaguar trade included quantitative information that would enable an assessment of the number of jaguars harvested. Purpose refers to the motivations behind the trade, and whether they were driven by profit (commercial) or by other factors like culture, self-defence, subsistence. Location refers to physical location of the events and whether they occurred in-country or abroad. Time pertains to the date of the events

Card	Nationality	Source	Scale	Purpose	Location	Time
An international magazine recently published photos of an indigenous festival that occurs annually in a neighbouring country, which showed three indigenous leaders wearing jaguar skins and jewellery made with jaguar fangs	Local	Media	Quantitative	Non-commercial	Foreign	Current
A friend from childhood told you that, in the countryside where he has a farm, poor villagers who depend on agriculture and subsistence hunting for protein, often encounter jaguars during their hunts and kill them in order to consume their meat	Local	Acquaintance	Undefined	Non-commercial	Local	Current
A trusted colleague from work, who lives in your same city, told you that he currently knows four people in his neighbourhood who own jaguar products for decoration, including jaguar skins, teeth and claws	Local	Acquaintance	Quantitative	Undefined	Local	Current
An academic study conducted 2 years ago by a researcher from a foreign university, who interviewed 35 ranchers and farmers in rural villages across the country, found that four of the respondents (11% of the sample) had been involved in the trade of jaguar skins, teeth and claws	Local	Research	Quantitative	Commercial	Local	Past
A conservation NGO surveyed 50 cattle ranchers in the nearby countryside and found that 10 ranchers (20% of the sample) had killed a jaguar within the past 5 years in retaliation for cattle losses. Upon questioning, the ranchers confessed that they had retained the skins of the jaguars as trophies	Local	Research	Quantitative	Non-commercial	Local	Past
A researcher from a local university conducted an online trade analysis focusing on illegal wildlife trade in the country and found that a few months ago, 10 people were advertising jaguar products, including skins, teeth and claws, on Facebook and other social media	Local	Research	Quantitative	Commercial	Local	Current
A reputable local newspaper reported that within the Chinese communities in this country, jaguar bone is currently being used in replacement of tiger bone in medicine used to treat arthritis and rheumatism	Foreign	Media	Undefined	Undefined	Local	Current
A study that was recently published by a conservation NGO in a neighbouring country revealed that 60 jaguar fangs (equivalent to about 15 jaguars) have been seized across the border of this country	Foreign	Research	Quantitative	Commercial	Foreign	Current

(Continues)

TABLE 1 (Continued)

Card	Nationality	Source	Scale	Purpose	Location	Time
The Minister of Tourism just announced the seizure of 60 jaguar fangs (equivalent to about 15 jaguars) at the city's airport. The fangs were being transported by a group of European tourists, who visited local markets in the city and purchased the fangs as souvenirs	Foreign	Government	Quantitative	Commercial	Local	Current
The local police recently found 12 jaguar fangs (equivalent to about three jaguars) at the residence of a Chinese citizen who migrated to this city a few years ago	Foreign	Government	Quantitative	Undefined	Local	Current

discussion with interviewees. We also tested whether interviewees' characteristics (nationality, education level, gender, organization or profession) influenced their choice of 'explicit criteria' in the discussion of the card prioritization, through single attribute logistic Bayesian generalized linear models, implemented through function 'bayesglm' in the `ARM` R package (Gelman & Su, 2020).

3 | RESULTS

3.1 | Factors influencing decision-making processes

Interviewees discussed the processes and factors involved in decision-making surrounding IWT within their institutions (Table 2). Most (78% of 38) interviewees said that their work is highly reactive; being largely focused on addressing emerging needs and threats originating from the interactions between people, protected areas and wildlife. Many of those emerging needs and threats have a temporal component, fluctuating with the reproductive stages of particular species or with weather-dependent accessibility to wildlife habitat (for both rangers and poachers). When addressing these emerging needs, interviewees also considered the safety of their staff and equipment (e.g. camera traps), due to the presence of armed poachers. Sixty per cent of our interviewees mentioned that their IWT actions depend on emerging funding or partnership opportunities to start new projects or to sustain ongoing initiatives. Meanwhile, 55% said that they follow pre-established organizational agendas, focused on targeting key vulnerable species or areas covered under the institutional mandate, or on providing inputs for policy. Forty-eight per cent of interviewees stated that their decision-making follows research recommendations, while personal experiences and interests were mentioned by 20%.

3.2 | Commonly consulted IWT evidence sources

Interviewees discussed the sources of evidence that they commonly use for decision-making surrounding IWT topics (Table 3). Most (73% of 38) interviewees said that they rely on data collected through research and monitoring activities. These included site patrols monitoring illegal activity, biological and ecological studies, studies on community livelihoods and socioeconomic indicators, as well as national-scale surveys on wildlife use. Sixty per cent said that they base their decision-making and daily actions on information received from the public, especially social media posts or phone calls reporting illegal activities concerning wildlife. Conservation institutions were another important source of information for 58% of our interviewees, many of whom were part of institutional working groups on matters related to IWT in their countries and used their periodic meetings to share information, while others described more informal communication channels among institutions such as WhatsApp groups. Forty-five per cent of our interviewees worked closely with local communities, and received information on illegal activities from

TABLE 2 Factors influencing decision-making. Categories add up to more than 100% due to non-exclusiveness. Sample size = 38

Factor	% of interviewees	Key quotes
Threats and needs	78%	'When it comes to prioritising, we focus on what is most urgent and a lot of the times it's either enforcement or conflict issues because sometimes we get a call, "if you don't come today, that jaguar dies tonight". Sometimes it is that serious and that's when we have to prioritise that. It's not that we don't have priorities, but we have to live day by day' (I20)
Funding and Partnership Opportunities	60%	'Our decisions are dependent on funding. When I was working with jaguars, we only had two months' worth of funding to focus on the species, so then I moved on to macaws because we won new funding for a project. Then we wrote a proposal to focus on communities, and so on' (I38)
Organizational agenda	55%	'In our institution we have a manual of procedures with the lines of action of our department with regards to wildlife. Sometimes we have the willingness to address the trade of a certain species like jaguars, but we do not have that activity in our annual or monthly operative plan, and so we cannot act' (I12)
Research and monitoring	48%	'When I began to work with the issues of wildlife trafficking, there was not a lot known... We managed to do a few studies to gather some data of what was the situation and which were the main species being trafficked. Along that path we had a few incidents which made us realise that the jaguar was one of the species needing to be considered' (I15)
Personal experience and interests	20%	'It's complicated because in the end all priorities depend on the interest of each person. I leave all administrative work for last, and answer to requests from public institutions or courts first, or rescue wildlife first. Those are my priorities because we are dealing with living beings' (I36)

them. Information from government agencies, particularly from enforcement operations carried out by the police or military, was another important IWT evidence source (40%). Meanwhile, 10% used information from the media or from personal experience (5%).

3.3 | Evidence prioritization

While participants described using information from multiple sources, not all evidence is considered equally. The card ranking exercise revealed how the different attributes encoded in the cards (time, space, nationality, purpose, scale and source) play a role in evidence prioritization. Taken together as a group, interviewees recognized and referred to all of the evidence attributes that we hypothesized as being important for evidence evaluation when independently describing their prioritization process (e.g. explicit criteria). However, not all interviewees noticed all attributes encoded in the cards without being prompted. The criterion that most interviewees independently noticed and used for prioritizing the evidence was the purpose or motivation behind the trade (68%), followed by how long ago the event took place (time, 58%), the nationality of the offender (45%), the location of the events (32%), the scale or number of body parts or traders (32%) and the source of the information (29%, Figure 1).

When prompted to discuss the evidence-prioritizing attributes in more detail (including those that they did not initially notice on

their own), interviewees revealed some of their opinions behind them. Concerning the purpose of the trade, the majority of our interviewees (89% of 38) mentioned that non-commercial motivations behind the killing and trading of jaguars, such as cultural, subsistence or retaliation-related uses, are 'understandable' because they involve longstanding traditional practices, they emerge from the urgent needs of vulnerable communities, and they are 'probably sustainable'. Therefore, these non-commercial purposes or motivations would warrant a lower priority level than commercial purposes, which were regarded as more detrimental and deserving of more attention, as stated by 42% of our interviewees.

All interviewees agreed that events happening in real time are a priority, because they require immediate action that can lead to an arrest or a seizure, which would allow them to demonstrate their ability to respond effectively to IWT, a key factor for their job security. These took priority over past events, because according to interviewees, the chances of achieving results diminish with time. The importance of the timeliness of the information also meant that information from studies that took place in the past was less valuable than anecdotal accounts on current events, regardless of the differences in the quality of the information.

The nationality of the traders was an important attribute for interviewees, and 82% of them specifically believed that foreign traders deserve a greater focus, particularly traders of Chinese descent. This concern was explicitly related to the impact of Chinese demand on tiger populations in Asia, as stated by 26% of our interviewees.

TABLE 3 Information sources used in decision-making. Categories add up to more than 100% due to non-exclusiveness. Sample size = 38

Sources	% of interviewees	Key quotes
Research and monitoring	73%	'We have conducted several studies, some based on interviews with hunters and local communities, some with camera trapping, and we have also done studies on DNA samples. So based on that scientific data we decide what to do and what the priorities are' (I21)
The public	60%	'We get the data from the public. It is mostly a response to information that arrives from calls of people seeing illegal activities, or from activism groups, or because the neighbour told a friend, or the chairman of the community, or even coming from outside the community' (I8)
Conservation institutions	58%	'Whichever member of the working group encounters wildlife trade, they share the information communally within the group. Sometimes we get reports about a demand for body parts of jaguars and other wildlife, if there's been any incidents of them being kept in captivity, or lethal measures for removing the jaguars and that sort of thing. Because there are several organizations, you get a lot more information about what is going on the ground' (I29)
Local communities	45%	'I work with around 300 people who are local informants. They are people that are worried about the damages that people cause to natural resources, and especially to animals. It's a network of people from local communities that look for information for us, because otherwise, if you are a stranger, people won't say anything' (I27)
Authorities	40%	'I work a lot with police officers and the military. If the police is doing whatever activity and they come around wildlife they will give us the information, and then we would do a little investigation and so on. The military does have an intelligence unit and they have officers that are not in uniform, so I would ask them to do a little investigation for us and they would go into the area and ask around and report back to us' (I37)
Media	10%	'Asian demand is a problem and the topic of jaguars is starting to arrive to this country. We have heard it in the news. We know that it is coming from South America and that they are moving in this direction' (I38)
Personal experience	5%	'There is very little information published in this country about wildlife trafficking, all the information we have is based on our experiences, what we have seen, what we have heard, what we have seized' (I25)

With regard to the location, for 53% of interviewees, the highest priority went to events occurring within their country and/or area of jurisdiction. As with time, the spatial proximity of events affected interviewees' ability to intervene, as trade happening

across the border fell outside their mandate. 77% of interviewees described how, while international and transboundary trade is important, particularly if it concerns neighbouring countries which share similar IWT routes and trade chains, addressing

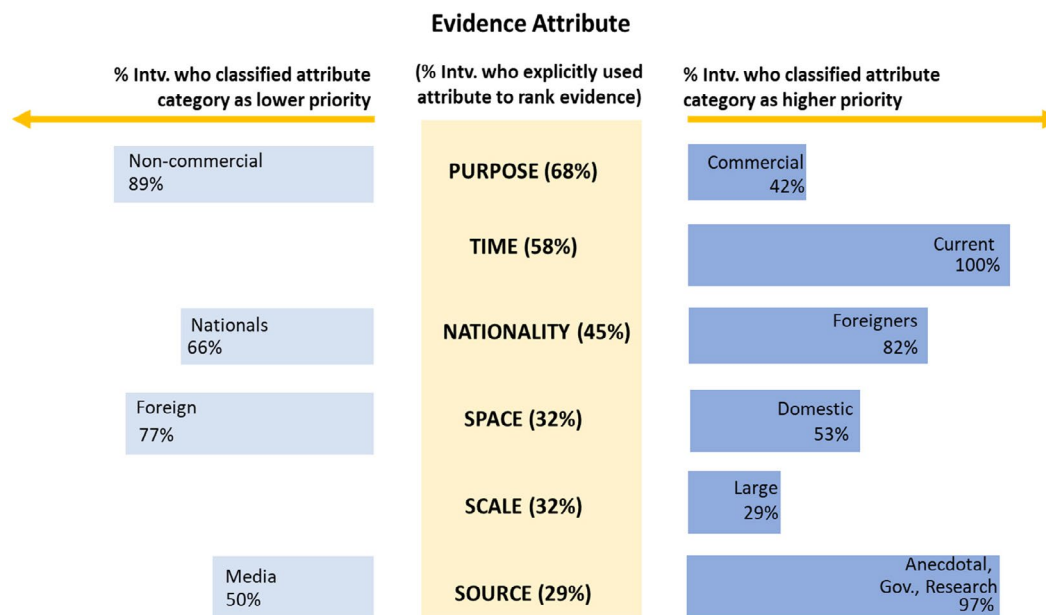


FIGURE 1 Percentage of interviewees ($n = 38$) who explicitly used the different attributes (purpose, time, trade nationality, space, scale and source; middle yellow bar) to prioritize evidence. Attribute categories were coded according to whether they were responsible for higher (darker blue horizontal boxes) or lower (lighter blue horizontal boxes) prioritization of a piece of evidence (Table 1) when prompted. Attributes and their categories are not mutually exclusive and may sum to more than 100%

these events requires strong partnerships, which are not always in place.

Although not all interviewees noticed the quantities encoded into the cards or prioritized the cards in terms of the numbers of jaguars affected or traders involved, when prompted, 29% mentioned that larger scale trade mattered most. The scale of the trade was closely associated with the nationality of the trader and purpose of the trade. For example, events related to foreign demand or to commercial purposes were perceived by 21% of interviewees as representing a larger potential impact on jaguars, involving large markets and high prices that could ultimately affect jaguar population sizes, while non-commercial uses were perceived to happen at small scales only (as suggested by 45% of interviewees).

Even though the source of the evidence was the least important attribute for interviewees, based on their explicit prioritization discussion, when prompting them to speak more about the source of the evidence interviewees mentioned that anecdotal accounts, government reports and research information are all important sources. This was in contrast to the media, which was described by 50% of interviewees as 'sensationalist' or 'unreliable'. Five interviewees shared personal experiences of being misquoted or misinterpreted by journalists. The most important sources of information for participants were their networks of friends and colleagues and social media (82%). This was not so much related to the reliability of the information, as 45% of interviewees mentioned that anecdotal information requires additional work to validate, but because these sources were perceived as being more timely and directly actionable by themselves personally. Information from interviewees' network

was also described as being trustworthy. Official sources like the government or research carried out by NGOs or academics was considered highly valuable by 47% and 39% of participants, respectively, but 18% expressed concerns about official sources being unreliable as they respond to institutional agendas.

Interviewees' characteristics, such as gender, profession, affiliation, education or nationality, did not influence how the evidence was explicitly prioritized, except that men were more likely than women to focus on the nationality of the traders ($\beta = 2.65$; $CI = 0.74-4.57$; $p = 0.007$) and less likely to notice the source of the evidence ($\beta = -1.78$; $CI = -3.33-(-0.21)$; $p = 0.03$, Appendix 1).

The explicit criteria that interviewees used in prioritizing the evidence, and the detailed discussion about the evidence attributes, differed from the implicit attribute ranking based on the physical card arrangement. Whereas interviewees explicitly prioritized the purpose of the trade and the timeliness of the information (Figure 1), the physical card arrangement revealed that higher ranking cards were significantly more likely to be those which described foreign traders, and events happening at the national level (Figure 2). This shows that trader nationality and the location of the events were more implicitly important than interviewees admitted to during the discussion. However, the direction of the effect aligns with the discussion, confirming that concerns over the role of foreign traders supersedes those related to local traders, and that domestic events take priority over transboundary or international ones. Other attributes encoded into the cards, such as the scale, purpose or time, did not influence the card ranking significantly, with the exception of the source. In particular, evidence coming from acquaintances ranked

higher than those originating in other sources (e.g. research, government), but only media sources led to a significantly lower ranking. This also coincides with the discussion, highlighting the important value of anecdotal information to interviewees, particularly when it comes from within a trusted network of friends and colleagues.

Based on both types of evidence prioritization, we were able to find support for our hypotheses that local and current events are prioritized over distant and past ones, as are foreign-driven commercial events over domestic and non-commercial ones, partly due to their perceived impact on the scale of the trade. Contrary to our hypothesis, official sources were not prioritized over anecdotal ones, and interviewees appeared to prefer direct and actionable over high quality information.

3.4 | Reactions to the evidence

Several interviewees explicitly stated that ultimately what matters to them is whether the information is plausible given their own understanding of the local context (e.g. hypotheticality, 42%) and whether the information lies within their specific mandate (34%). Interviewees stated that evidence that did not match their contextual understanding of their areas, or which extends beyond their duties would simply lead to concern (24%) or inaction (21%). For jaguar trade evidence that is considered relevant, interviewees' most common stated potential reactions would be to develop projects to address the trade at the community level (76%), to investigate (66%), to enhance response capacity by searching for funds and building partnerships (55%) and to conduct enforcement operations (45%).

As described in Arias et al. (2020), the main challenges to carrying out these actions for the specific case of jaguar trade include the lack of financial, human and physical resources (73%, $n = 41$,

including responses of three additional interviewees who did not participate in the evidence evaluation), the ineffectiveness of law enforcement systems (68%), animosities between actors (54%), corruption (51%), outdated legal systems (29%), lack of evidence (24%), lack of mandate to enforce the law (22%) and insecurity (17%).

4 | APPROACH FOR EVALUATING AND PRIORITIZING IWT EVIDENCE ACTION

Our discussions about decision-making and evidence use with conservation practitioners working to address illegal jaguar trade in Mesoamerica provided valuable insights into some of the challenges of implementing evidence-based conservation. Even though interviewees explicitly stated that they use scientific evidence in their decision-making processes, they were susceptible to the implicit attribute configurations of the evidence. In general, interviewees gave more weight to evidence involving foreign actors and commercial purposes than local and non-commercial ones, and to events closer to their reality in space and time, over more distant ones, paying less attention to their implied impact on jaguars or the validity of the source of the information. These biases do not mean that conservation practitioners are not driven by evidence, but rather that the definitions and interpretations of evidence are subjective, even amongst scientifically trained and experienced decision makers.

We propose an approach for evaluating evidence and informing decision-making within IWT and biodiversity conservation more broadly (Figure 3). Our approach can support IWT decision makers and practitioners to judge when there is enough ground to invest in an IWT response (e.g. increasing enforcement or implementing behaviour change campaigns); when the inconclusiveness, uncertainty or irrelevance of the evidence merits continued research or

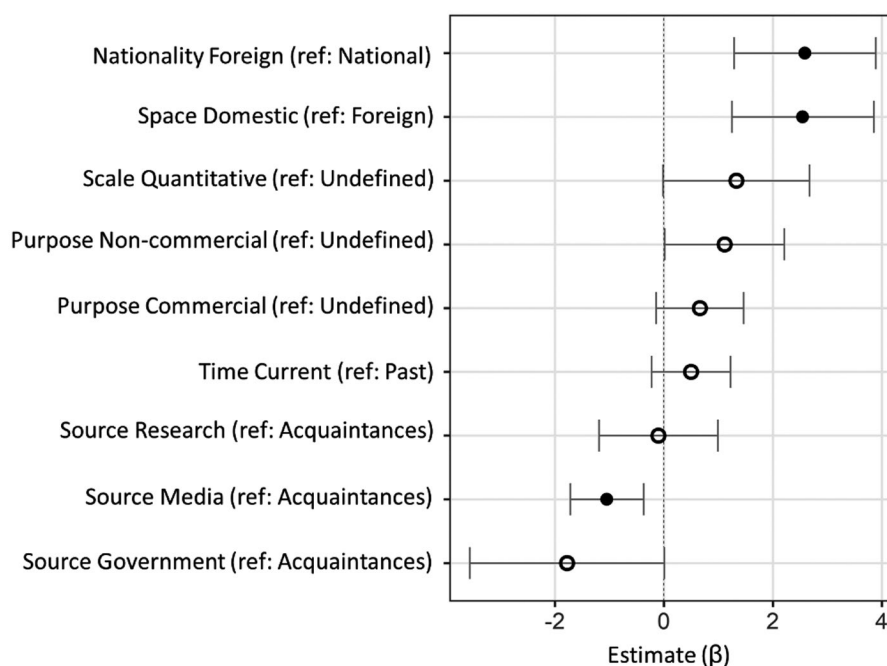


FIGURE 2 Implicit evidence prioritization by interviewees ($n = 38$) based on physical card arrangement. Estimates of a cumulative link mixed model with 95% Wald confidence intervals. Solid circles represent significant effects (p -value = 0.05). Ref: the reference level for a factor

monitoring efforts, and when to safely dismiss IWT evidence in order to centre efforts elsewhere. Our approach can be applied to any species threatened by IWT (and other threats) and it can be used to evaluate single units of IWT evidence or a body of evidence.

Our evidence evaluation approach is based on four of the six evidence attributes that we used in this study, namely, the scale, source, time and location of the information. Even though the purpose of the trade and the nationality of the traders were among the main attributes considered by our interviewees for prioritizing jaguar trade evidence, we purposely did not include them in our evidence evaluation approach, as their importance should be assessed in light of the other attributes. For example, the purposes behind IWT or the nationality of offenders should only matter in relation to their potential impact on the species, and only if the evidence is reliable, timely and spatially relevant.

In our approach, evidence that merits immediate action or intervention is relevant in terms of its scale, source, time and space. Here, scale should be understood as the potential impact of illegal trade on the species of concern, as suggested by the evidence. Whether the scale or impact is considered high, medium or low will depend on several factors, including offtake quantities, species population numbers and life histories, as well as other bio-ecological and socio-economic factors which may influence the level of threat posed by trade to the species, in a given space and time (Bennett et al., 2021). These can be assessed qualitatively or quantitatively through expert elicitation or through models of varying complexity, parameterized with as little or as much data as are available, reflecting the uncertainty around the predicted impact (e.g. Weinbaum et al., 2013). If resources are limited and the scale or impact of trade on a species is determined to be low, such that it does not compromise its current conservation status, the trade evidence is unlikely to justify an intervention other than continued research and monitoring to determine whether the impact is likely to grow in the short, medium or long term. If it is unlikely that the impact will grow in the foreseeable future, the threat merits monitoring over time to detect any changes, after which it can be safely be dismissed.

To become an intervention priority, the evidence must also be reliable. Reliability is a measure of bias minimization and it is often assessed against the steps taken to reduce common biases in experimental design, such as selection, performance, measurement or attrition bias (Cooke et al., 2017). Drawing from the medical sciences, high levels of reliability or certainty are often assigned to randomized control experiments, and decrease for observational or qualitative studies (Morgan et al., 2016). However, considering that IWT evidence rarely comes from experimental studies, we suggest that the trustworthiness of the source can be considered a proxy of reliability. Evidence originating from official sources, such as enforcement reports or academic studies, may be considered more reliable than anecdotal accounts or information from non-official sources. We recognize that this will vary strongly with the context (e.g. even peer-reviewed literature can be strongly biased, Cooke et al., 2017), and that other variables should also be considered (e.g. legitimacy, Cash et al., 2003). It is therefore up to the assessor to

justify the classification of a source as reliable or unreliable, and to be accountable for that decision. If the source is not deemed reliable, the evidence deserves further research and monitoring to establish whether the issue or hypothesis can be reliably proven with more evidence. If not, monitoring efforts should take place periodically to determine whether new evidence has surfaced, followed by dismissal of the evidence if it cannot be corroborated.

Whether or not the evidence is current (time) and local (space) also influences its relevance for intervention. IWT is highly dynamic and quickly adapts and responds to new market trends, changing prices and enforcement efforts. Although time-scales are context-specific, outdated evidence can lead to inappropriate IWT responses. Before investing in actions to address IWT, decision makers should determine whether the evidence is an accurate representation of the current reality, and if it is not, the research priority should be to assess its representativeness or the likelihood of recurring trade incidents. Recurrence of IWT should be monitored over time, and the evidence can be safely dismissed if it is found to be untimely or no longer representative. Similarly, evidence that is not pertinent to the geographical jurisdiction and range of action of those in charge of the IWT response is unlikely to warrant its classification as a priority for action. However, the potential for IWT to spread should flag it as a research priority. Even if IWT does not appear to be spreading, changes in its movement (e.g. the reach of its trade chain networks) and density (e.g. trade hotspots) should be monitored over time, until the evidence can be safely dismissed.

Not all evidence attributes need be at their highest level (green area in Figure 3a) to warrant an intervention. A single piece of evidence can combine attributes at different levels (Figure 3b), and it will be up to decision makers to assess whether it is the right moment to intervene, assuming that resources and capacity are available, or whether it might be more appropriate to delay an intervention. Rather than providing prescriptive cutoffs for action, our approach can serve as a tool to reflect upon the attributes of the evidence, and to facilitate transparency behind decision-making. For example, if all evidence attributes are judged to be at their highest level of relevance (green area in Figure 3a), it would be difficult to justify inaction. On the contrary, if all evidence attributes are at their lowest level of relevance (red area in Figure 3a), any efforts aside from research or monitoring could run the risk of being unjustified or not based on good evidence. Evidence combining low-, high- and middle-level relevance attributes can be more complicated to decide upon. In these cases, decision makers can rely on research and monitoring to determine whether any of the attributes have changed and to increase confidence in their decisions. For all attributes, just as the evidence can move from intervention priority to dismissal depending on its evolving characteristics, it can move in the opposite direction following changes at the monitoring or research stages.

A common intervention used in counter-IWT projects across the world is to provide training about IWT to law enforcement agents, and wildlife or justice authorities. This training, which is generally conducted by experts or NGOs, could offer an opportunity to discuss the issues around IWT evidence, and to introduce our approach

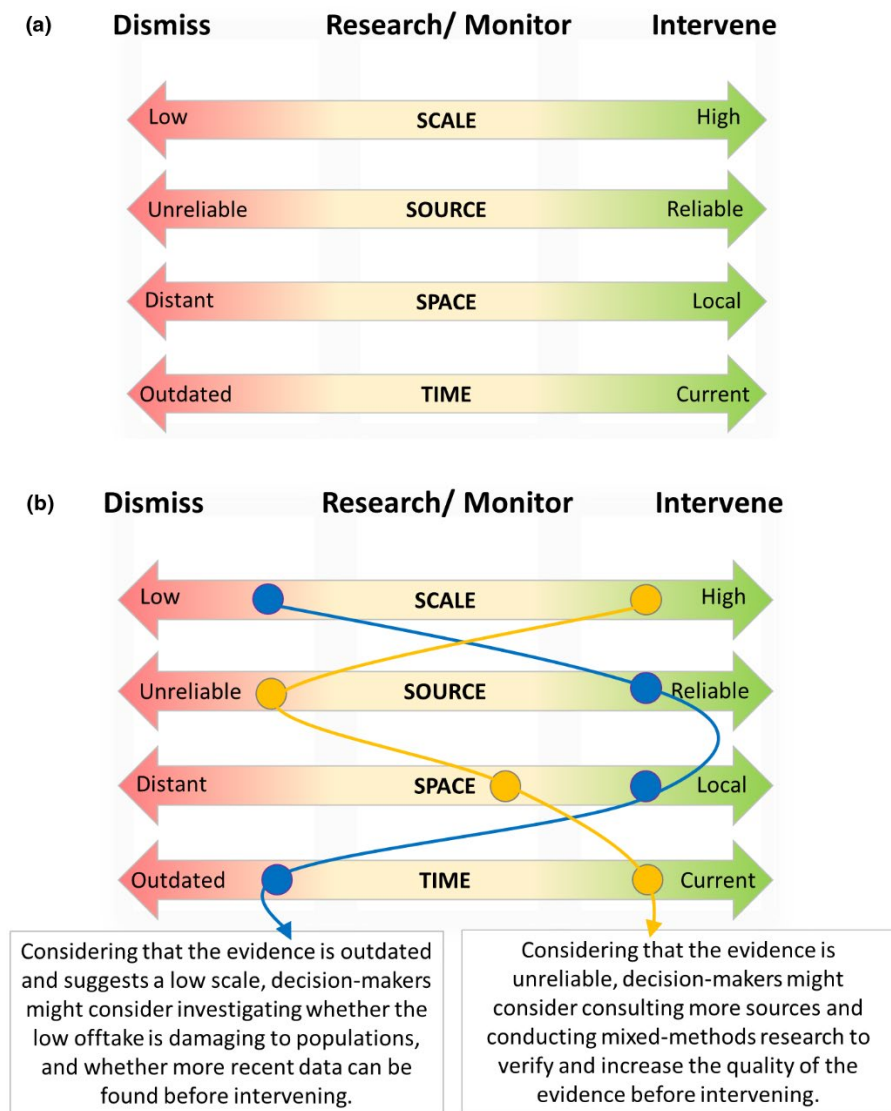


FIGURE 3 (Panel a) Approach for assessing illegal wildlife trade evidence priority. Variation in the scale of the trade or impact on populations suggested by the evidence (high to low), its reliability (reliable to unreliable), space (local or distant) and time (current or outdated), in no particular order, determine whether there is enough ground to intervene to address the trade, when to prioritize research and monitoring instead, and when it is possible to safely dismiss evidence. (Panel b) Hypothetical examples of approach implementation with combinations of evidence attributes. Yellow and blue circles represent two pieces of evidence with different attribute configurations

to conservation decision makers and practitioners. Similarly, an explicit consideration of the four evidence attributes in our approach, or more broadly about evidence biases and uncertainties, could become a requirement in IWT-related grant applications, IWT project authorization processes, and in national and international risk evaluations for species in trade (e.g. national or IUCN Red List listings; CITES non-detriment findings or reviews of significant trade).

While our approach is tailored to IWT and conservation decision makers and practitioners, it can also be of value to conservation scientists, practitioners, journalists and any stakeholder interested in producing evidence to support decision-making in biodiversity conservation. The same evidence attributes that combine to justify policies and conservation interventions (Figure 3) can also guide

researchers and other evidence 'producers' to increase the relevance of their work. In choosing their research questions or in their dissemination of results, evidence producers would benefit from exploring the evidentiary weight and the evidence attribute configurations behind their chosen angle or positionality towards the issue. They should further reflect on the evidence underpinning their perspective compared to other perspectives, and whether they can identify any key assumptions or potential biases in their own views that should be addressed or disclosed (e.g. on the nationality or purpose of the trade). Early on, evidence producers should also identify and learn about their intended audiences (e.g. wildlife authorities, business leaders) and ideally include them in the evidence production process. They can then emphasize the attributes that

make their evidence inputs spatially and temporally relevant to their target audience, actionable (within their audience's mandate), meaningful in scale or scope and to communicate their work strategically in ways that foster trust and which emphasize the reliability of the information. Considering that the timeliness and actionability of the information is a key consideration for conservation decision makers and practitioners working to address the IWT, researchers and other evidence 'producers' should consider sharing their confirmed findings earlier on in the publication process (or even beforehand), respecting their ethical responsibilities to informants.

A vast interdisciplinary literature has provided recommendations for facilitating the understanding of, and engagement with, evidence recipients to increase the uptake of evidence. Some of these recommendations include opening spaces for personal interactions with people across disciplines and sectors; co-production of research from the inception of projects; taking an interest in the governance systems and culture behind decision-making; communicating in more appealing ways in light of the role of empathy and emotion in decision-making; understanding people's natural tendencies to misunderstand risk, uncertainty and complexity; and supporting knowledge brokers (Cvitanovic et al., 2016; Gore et al., 2020; Mercier & Sperber, 2017; Oliver & Cairney, 2019).

5 | DISCUSSION

Given the vast uncertainty in IWT, decision makers face the difficult task of sifting through evidence of varying types and qualities to identify priorities for action and investment. Regardless of their good intentions, they face multiple challenges to conducting evidence-based conservation, from managing the uncertainty in the evidence to navigating the complex governance processes influencing evidence communication, interpretation, socialization and uptake (Bertuol-Garcia et al., 2018; Nutley et al., 2012; Walsh et al., 2019). To deal with this complexity, conservation decision makers and practitioners may opt for 'rational' shortcuts, such as putting their trust in experts and scientific sources, or 'irrational' shortcuts, relying instead on their own or others' beliefs, emotions, habits and familiarity with the issues and potential solutions (Oliver & Cairney, 2019; Sloman & Fernbach, 2017). For complex and uncertain topics like IWT, priorities may end up being established on the basis of assumption, lobbying or on the grounds of dominant IWT narratives rather than on scientific evidence (e.g. Margulies et al., 2019). This may lead to ineffective interventions that fail to target the actual drivers of IWT, or to the dismissal of topics and taxonomic groups that fail to capture policy interest (e.g. 'plant blindness').

We propose an approach that can help decision makers to distinguish between actionable evidence and evidence of lesser value to decision-making, on the grounds of its temporal and spatial relevance, reliability and suggested impact on wildlife populations. While our approach does not solve all the challenges of evidence-based decision-making on IWT, it encourages a critical assessment of evidence, including the management of uncertainty. Being based

on the evidence prioritization process of real conservation practitioners and decision makers working to address IWT on the ground, our approach highlights the attributes that matter most when prioritizing evidence in real-world contexts. It also aims to promote self-reflection amongst decision makers about their own implicit and explicit biases, and to help them to identify pieces of evidence, or specific attributes in the evidence, that are likely to receive an unjustified weight. Our approach can also increase the transparency and accountability of decision makers' choice to take a given course of action based on the evidence, from intervening to address IWT, to commissioning research on a new threat, to monitoring IWT threat levels over time and to the safe dismissal of evidence. Our approach would benefit conservation practitioners making day-to-day decisions to address IWT at specific sites, just as much as policymakers working on IWT at national or international levels. It can also support scientists, journalists and other evidence 'producers' to build evidence that is more oriented towards conservation policies or actions, and to communicate it in a timely manner and in appealing ways that foster trust and reliability.

Researchers at the science-policy interface have called for 'case studies of different types of policymaking and the evidence diets consumed in these contexts; and how different people embody different imperatives of the evidence/policy nexus' (Oliver & Boaz, 2019). Our interviews with practitioners and decision makers working on illegal jaguar trade, and other types of IWT in Mesoamerica, elucidated some of the practicalities of using evidence for decision-making on IWT and biodiversity conservation. We found that decision-making in the region is highly reactive, responding to urgent needs and threats that arise as species become threatened, and to emerging funding opportunities. In this context, the most valuable evidence for practitioners is not necessarily the most robust or ecologically pressing, but the one that draws attention, collaborations and funding, and for those working on the ground, the one with a potential to reach a tangible outcome, like a seizure or arrest. The use of evidence is therefore closely related to the incentives that are in place for decision makers to engage with the evidence (e.g. seizing a high-profile species or gaining a funding grant), and to the specific implementation constraints that they face in carrying out their duties (e.g. Bertuol-Garcia et al., 2018; Cvitanovic et al., 2015). Similarly, interviewees recognized that long-term planning and evidence stemming from research and monitoring efforts are less likely to inform their action priorities when compared to more timely information from their networks, their own experiences and from non-official sources like the public, social media and the news. This diversity of sources aligns with the notion that conservation evidence can take multiple forms, and that decision-making in conservation is evidence-informed rather than evidence-based (Adams & Sandbrook, 2013). It also highlights the importance of making scientific evidence more relevant, timely, attractive and memorable to decision makers, something that can be achieved by engaging them in the evidence production process, building trust through lasting relationships, opening formal and informal channels of communication, using storytelling and a variety of dissemination

methods and enhancing their emotional connection to the evidence (Phoenix et al., 2019).

While there have been criticisms of decision makers' complacency about using evidence (Sutherland & Wordley, 2017), about scientists' passive science communication efforts (Cash et al., 2006) and about the lack of engagement on both sides (Cvitanovic et al., 2015), our results suggest that evidence use may also be hampered by decision makers' own implicit and explicit biases. Even though interviewees explicitly valued research, and many of them had a scientific training, they were influenced by implicit subjectivities and socially constructed pre-existing beliefs about IWT. Resistance to, or dismissal of, scientific evidence is common across all disciplines and sectors of society, and it is caused by complex psychological, behavioural, sociological, political and neurobiological factors that are intrinsic to human beings and vital components of our adaptive strategies for survival (Gorman & Gorman, 2016). Regardless of education or intelligence, humans are prone to adopting irrational behaviours due to natural tendencies to avoid complexity and uncertainty, to refuse change or departure from previously accepted beliefs, to misunderstand risks and to react in ways that prioritize emotion and empathy over science (Gorman & Gorman, 2016). Moreover, while human rationality performs well in social decision-making contexts, where it is used to argument and justify personal perspectives in contrast to those of others, our rationality is neither objective nor impartial in individual decision-making contexts. This is due to our overwhelming tendency to support what is already familiar to us and what is easier to justify (Mercier & Sperber, 2017). Additionally, we are all affected by the 'knowledge illusion', or the generalized failure in humans' cognitive ability to distinguish knowledge that is inside our own minds and knowledge that is embodied within the environment and in the collective thinking of our society (Sloman & Fernbach, 2017). Hence, our rationality and cognitive ability hinder our capacity to identify where our opinions come from and to recognize our own biases in our stance towards information. Considering these natural limitations, it is clear that experience and qualifications should not be regarded as automatic prerequisites for objective decision-making. Decision makers often fall victim to common biases such as anchoring (tendency to be influenced by initial information), availability (the influence of past experiences on memory), representativeness (single school of thought), groupthink and overconfidence (Burgman et al., 2011). Moreover, confirmation bias, or the inclination to retain (or disinclination to abandon) a favoured hypothesis, is ubiquitous in decision-making, and it affects how trained and untrained people search for, interact with and decide upon evidence (Busemeyer et al., 1995). Depending on their social environment, background and learning strategies, some people may be naturally more inclined to questioning their own understanding of an issue and using analytical processing to make sense of divergent information, while others may have a greater tendency to cluster around people and ideas that confirm their own and their group's understanding of an issue (Witting, 2017). Certain attributes in the evidence can trigger biases in how the information will be perceived, by evoking emotions and memories, and by positioning the content

in alignment with or opposition to decision makers' pre-existing beliefs (e.g. Newell et al., 2014; Toomey et al., 2017).

Therefore, overcoming the challenges to the use of evidence in conservation requires a deeper exploration of the cognitive science and psychology behind decision-making in conservation more generally. Conservation scientists can draw on multiple existing literature reviews across disciplines that have already achieved great progress in understanding the individual and social processes behind learning, evidence use and decision-making in policy contexts (e.g. Nutley et al., 2012; Oliver & Boaz, 2019; Witting, 2017). In particular, we recommend further research into the implicit and explicit biases in the use of conservation evidence by different people, under different contexts of data availability and uncertainty. Moreover, in alignment with the theory on policy learning and knowledge exchange, we encourage the opening of participatory and interdisciplinary spaces where evidence producers and evidence recipients can engage with each other and have honest discussions about evidence needs and uncertainty.

In the case of the illegal jaguar trade, fears about the role of China as a new market for jaguars have compelled several individuals and organizations across the jaguar range and abroad to act on the precautionary principle and to intervene to stop international illegal trade in jaguars. The justification is often to prevent jaguars from following the same path as tigers, which became endangered by Asian-driven IWT. However, the slowly growing evidence on the matter shows variations in how the trade is portrayed, from a predominantly international, Chinese demand- and corporate investment-driven (Lemieux & Bruschi, 2019; Morcatty et al., 2020; Verheij, 2019), organized trade (Romo, 2020; WAP, 2018), to a largely opportunistic and domestic issue, associated with cultural practices, human-wildlife conflict and tourism (Arias et al., 2020, 2021). Moreover, demand for jaguars in Asia is not yet understood, and the impacts of illegal trade on jaguar populations have not been empirically estimated. Due to this remaining uncertainty, research and monitoring efforts to determine the context-specific characteristics, drivers and impacts of the illegal jaguar trade should be a favoured course of action at the moment. Acting without enough evidence poses several risks that may frustrate the good intentions of the conservation community. For example, spreading information and raising awareness about jaguar trade could have the unintended effect of incentivizing it in areas where it currently does not exist. It could also spur a demand locally and internationally for objects that are perceived as trendy, endangered or rare (Hall et al., 2008). A lack of nuanced understanding of the trade could also damage relationships with local communities that are vital for wildlife conservation (Cooney et al., 2017; Duffy, 2014), spread prejudice about certain groups of people or even harm the economic relationship between jaguar range countries and China. Additionally, the already limited conservation funds and efforts focused on addressing human-jaguar conflict and habitat loss, two well-documented threats for jaguars (Castaño-Urbe et al., 2016; Olsoy et al., 2016), could be instead re-directed to a threat with less known impacts on the species. Moreover, classifying the jaguar

as the emblem of Latin America's fight against IWT could dismiss the trade in other wildlife, including timber, orchids, sharks, birds and reptiles, despite their large seizures and concerning population declines (UNODC, 2020).

IWT is a key concern for biodiversity conservation, and due to its illegal nature, it can be particularly hard to understand and address, especially for species that are recently affected by it. Decision makers at the forefront of IWT face the difficult task of undertaking timely actions to protect species while also dealing with the large uncertainty that is inherent to IWT evidence. Our interviews with decision makers working to address jaguar trade in Mesoamerica revealed some of the practical challenges of on-the-ground IWT decision-making, including how to prioritize evidence of varying types and qualities. Based on these insights, we propose an approach that can guide decision makers to recognize evidence of high relevance to conservation action from evidence of lesser quality that merits further research, monitoring or dismissal. We aim for this approach to contribute towards a more evidence-informed practice within the field of IWT, and biodiversity conservation more broadly.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interests.

AUTHORS' CONTRIBUTIONS

M.A. involved in conceptualization, methodology, formal analysis, investigation, data curation, writing original draft, visualization, writing—review & editing, project administration, funding acquisition; AH. involved in analysis, writing—review & editing, supervision. E.J.M.-G. involved in conceptualization, methodology, analysis, writing—review & editing, supervision.

DATA AVAILABILITY STATEMENT

This manuscript does not contain any interview data (transcripts) due to confidentiality.

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